

23/09/24

Routing Protocols

Routing Table

A routing table will store all the information about source ip address, destination ip address, subnet mask, hop, hop count, size of the packets and also the interface which connects one system to the other system.

Example

Active Routes:

Network Destination	Network	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.0.0.1	10.0.0.32	35
10.0.0.0	255.255.255.0	On-link	10.0.0.32	291
10.0.0.0-32	255.255.255.255	On-link	10.0.0.32	291

*In any routing algorithms 3 important parameters to consider are:

1) Forwarding the packets.

2) Updating the routing table (routing).

3) Delivery of the packets.

*Routing algorithms are classified into 2 types

1) Static algorithm

2) Dynamic algorithm.

DHCP - Dynamic Host Configuration Protocol.

Differences between Static & Dynamic Algorithm

Static

(Non-Adaptive)

1) Do not base their routing decision on any measurements (or) estimates of the current topology and traffic

2) The route to travel from source ~~to~~ destination is computed in advance

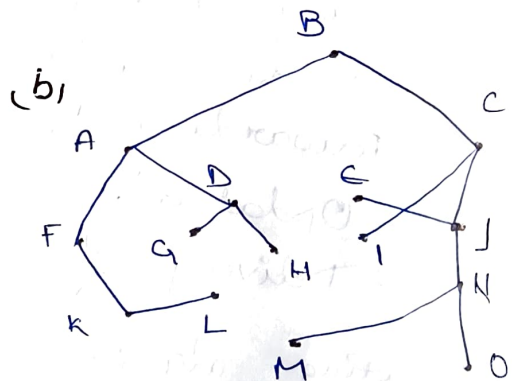
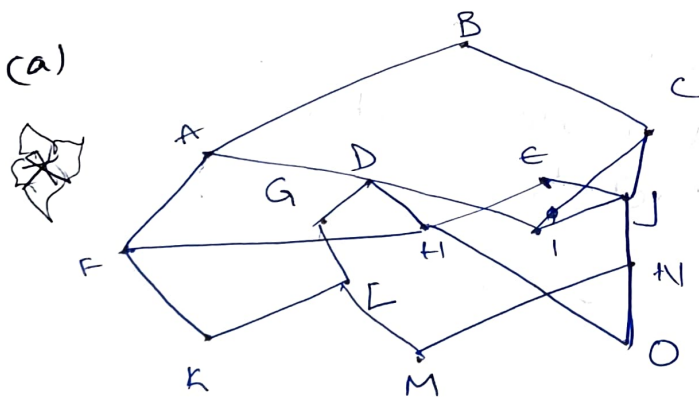
3) This procedure is sometimes called as static routing

Dynamic

(Adaptive)

1) change their routing decisions to reflect changes in the topology and sometimes changes in the traffic as well

The Optimality Principle



This is a sink tree.

=> A set of optimal routes from all sources to a given destination from a tree rooted at the destination.

=> This tree should not contain any loops, so each packet will be delivered within a finite hops.

Flooding

- * Every incoming to a node is sent out on every outgoing line except the one it arrived on.
- * It requires no network information like topologies, cost of different paths and load conditions.

Limitations:

- Flooding generates vast no. of duplicate packets.
- * ~~As~~ flooding generates vast no. of duplicate packets, an infinite number unless some measures are taken to damp the process (stop).

Hop:

- * A hop is a computer networking term that refers to the number of routers that a packet passes through from its source to destination.

Hop count:

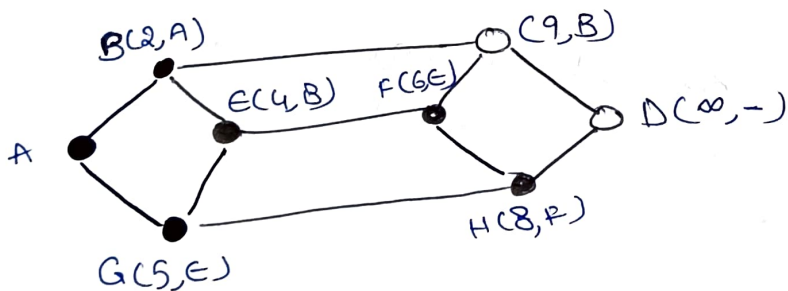
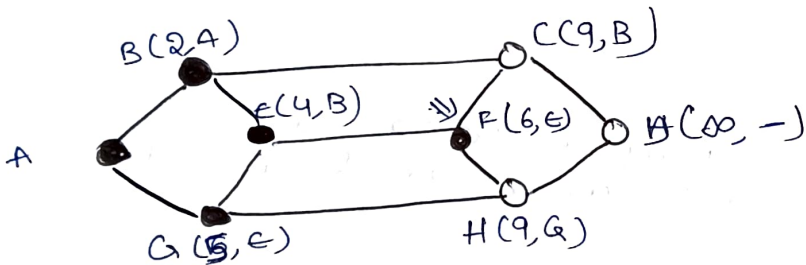
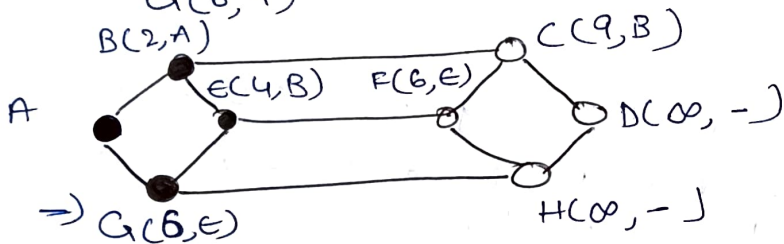
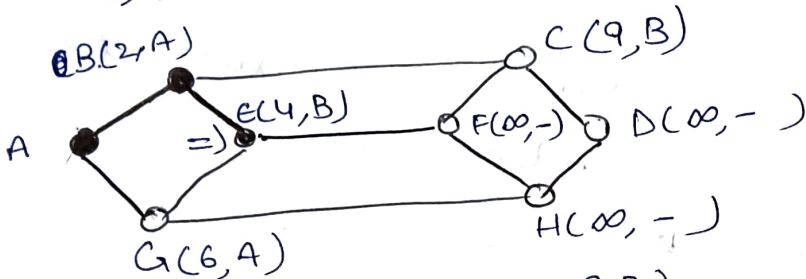
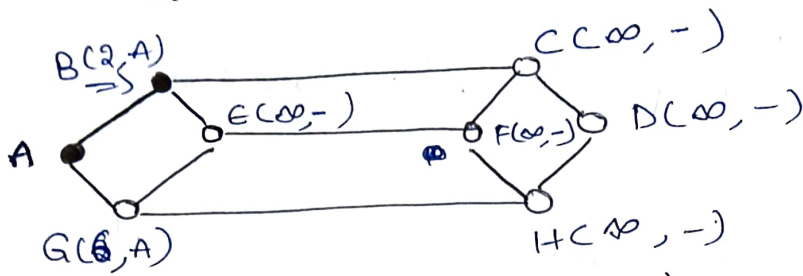
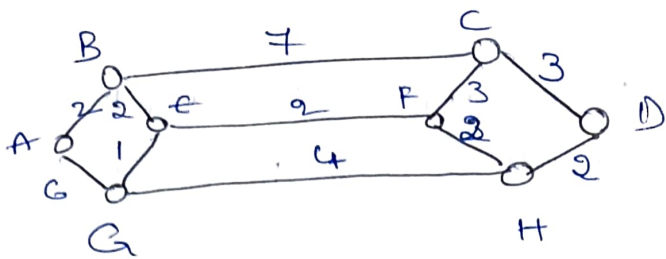
- * A hop counter may be contained in a packet header which is decremented at each hop.

Sequence Number:

- * Assigning each packet a sequence number to travel so that all the packets won't cause a collision.

Shortest Path Routing (Dijkstra's Algorithm)

- ⇒ Using Shortest Path algorithm find the best short path that exists between router A and router H.

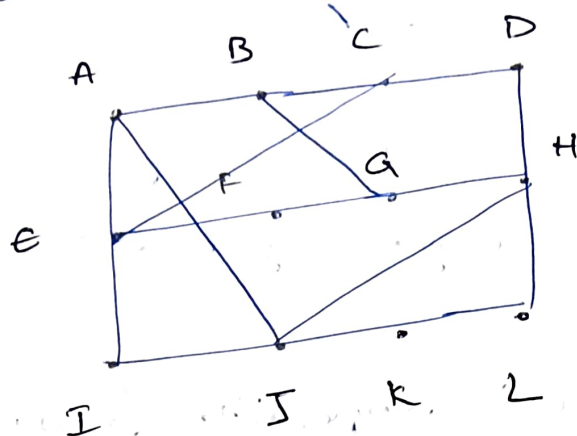


Distance Vector Routing

A distance vector routing algorithm operated by having each router maintain a table (i.e. a vector) giving the best-known distance to each destination and which link to use to get there.

These tables are updated by exchanging information with the neighbors. Eventually every

router knows the best link to reach each destination.



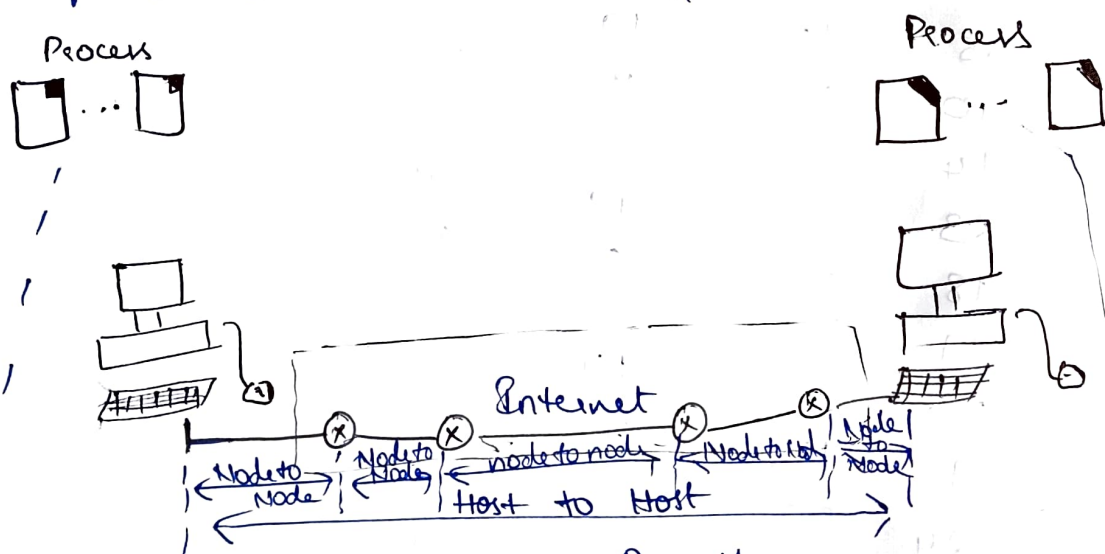
* If the packet needs to be delivered from routing point J the immediate routers and the tables to be updated are:

To	A	I	H	K
A	0	24	20	21
B	12	36	31	28
C	25	18	19	36
D	40	27	8	24
E	14	7	30	22
F	23	20	19	40
G	18	31	6	31
H	17	20	0	19
I	21	0	14	22
J	9	11	7	10
K	24	22	22	0
L	29	33	9	9

JA delay is 8
 JI delay is 10
 JH delay is 12

7/10/24 Transport layer : Process to Process
Delivery

- * Data link layer is responsible for delivery of frames between 2 neighbouring node over a link (node to node delivery)
- * The network layer is responsible for delivery of datagram between 2 hosts (host to host delivery)
- * Real communication takes place between two process (application programs)
- * The Transport layer is responsible for process to process delivery of a packet, part of a message, from one process to other.



Process to Process

Client - Server paradigm

A process on local Host - Client - Local process
needs services from a process on remote host - Server
remote process

* IP address decides between which hosts the data should be transferred.

* The port number decides between which process the data should be transferred.

IANA (Internet Assigned Numbers Authority) Ranges



Socket Address

It is an address which contains both IP address and port number.

IP: 200.23.56.8

Port number: 69

Socket address: 200.23.56.8 69

Multiplexing

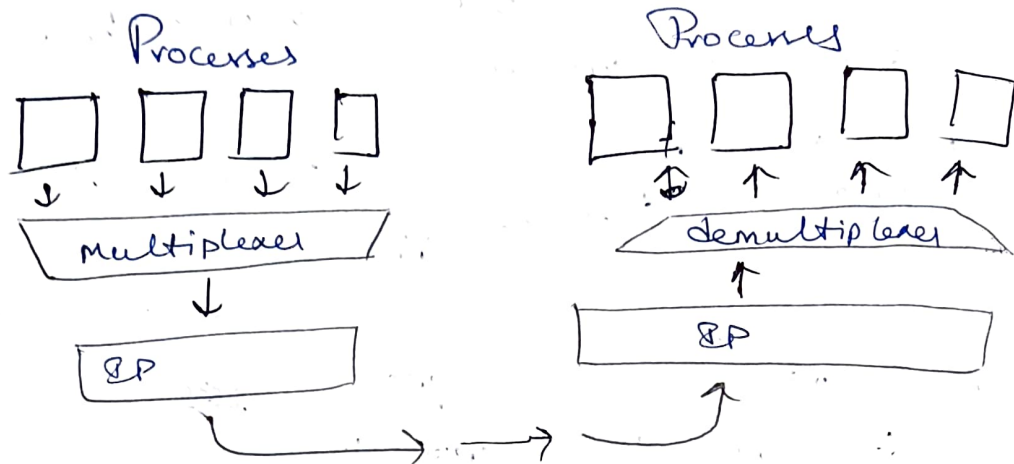
- At sender's site, there will be many processes that wish to send packets. However, there is only one transport layer protocol at any time. So it is a many to one relationship which requires a multiplexing.

Demultiplexing

- At receiver's site, there will be a one to many relationship and requires demultiplexing.

- The transport layer receives datagrams from the network layer. After error checking and dropping the header, the transport layer delivers each message to appropriate process based on port number.

With too much traffic, performance drops sharply.



* The transport layer can be either connectionless service (a) connection oriented service.

Connectionless service

In a connectionless service, the packets are sent from one party to another with no need for connection establishment, (a) connection release. The packets are not numbered, they may be delayed, lost (a) may arrive out of sequence. There is no acknowledgement either.

Connection-oriented service

* Here, the connection is first established between sender and receiver and then the data is transferred. At the end, the connection is released.

TCP (Transmission Control Protocol)

* It is a transport layer protocol.

* It is a process to process protocol.

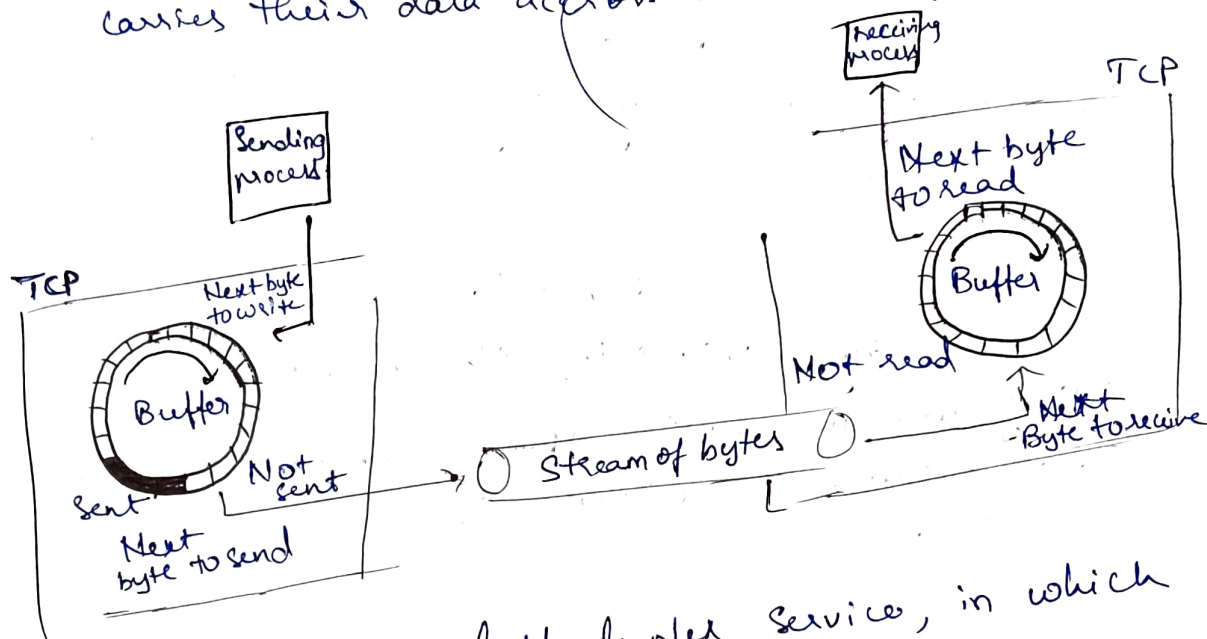
* It uses port numbers.

* It is a connection oriented protocol, it creates a virtual connection between 2 TCPs to send data.

* It uses error and flow control mechanisms.

Stream delivery service

- * TCP provides
- * It allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as stream of bytes.
- * TCP creates an environment in which two processes seem to be connected by an imaginary tube that carries their data across the internet.



- * TCP offers a full duplex service, in which data can flow in both directions at the same time. Each TCP then has a sending and receiving buffer and segments move in both directions.

- * TCP is a reliable ~~transport~~ transport protocol which uses an acknowledgement mechanism to check safe and sound arrival of data.

TCP features

Numbering System:

Byte Number: TCP numbers all data bytes that are transmitted in a connection.

Sequence Number: TCP assigns, a sequence number to each segment that is being sent.

Acknowledgement Number: Each party uses acknowledgement number to confirm that bytes it has received.

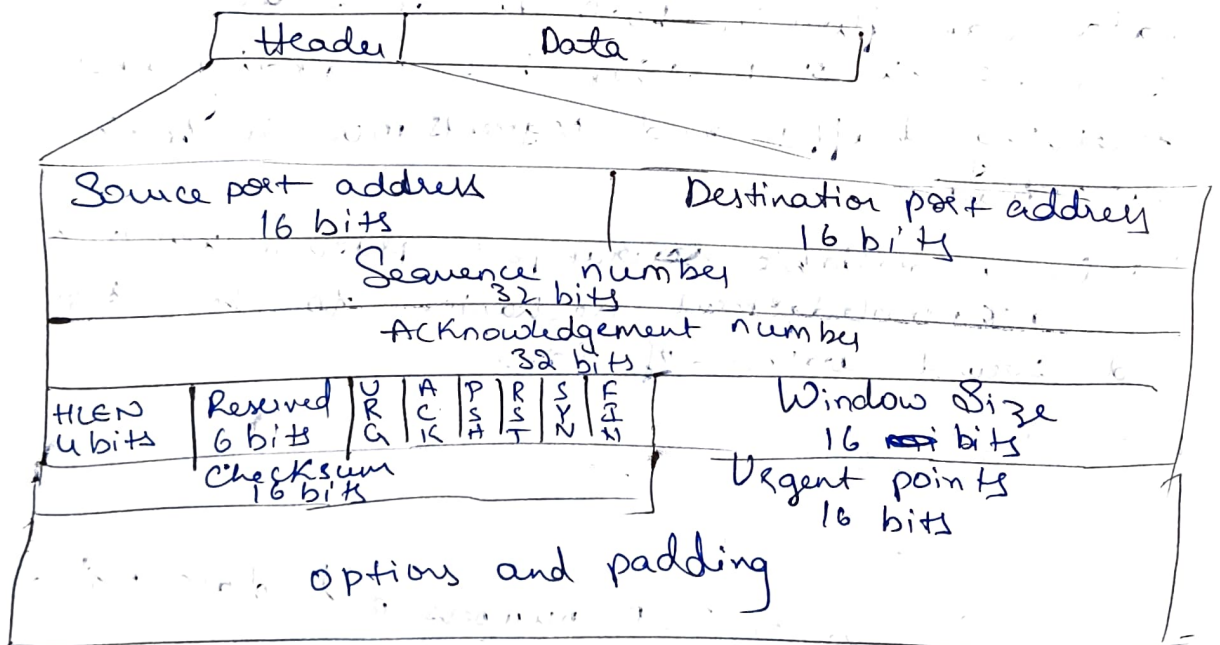
However, the acknowledgement number defines the number of the next byte that the party expects to receive.

Flow Control: TCP provides flow control. The receiver of data controls the amount of data that are to be sent by the sender.

Error Control: It provides error control for each segment of data by using checksum, Retransmission, Sequence numbers.

Congestion Control: It provides congestion control for data by minimising the network overload using window-sliding approach, Adaptive algorithm to reduce the window size when congestion is detected.

Segment



URG - Urgent

ACK - Acknowledgment

PUSH - Push

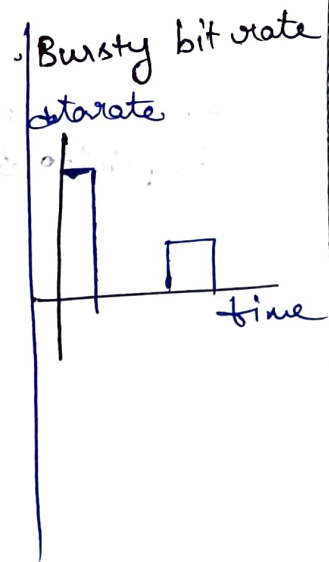
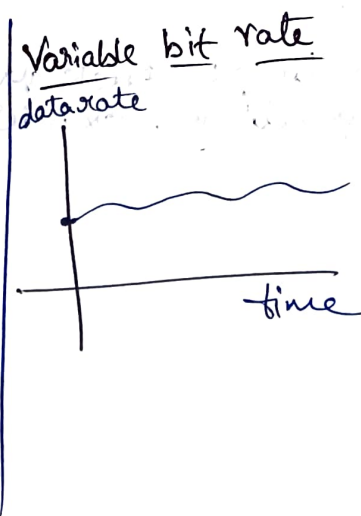
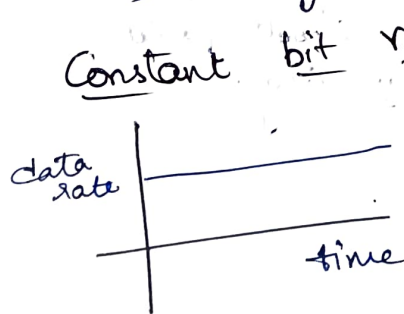
RESET - Reset

SYN - Synchronisation

FIN - Finish

15/10/24 Congestion Control

- * whenever segments, packets, bits are transmitting, there is always a problem of having data traffic.
- * Because of which the efficiency of that particular system will come down, by resulting in loss of information.
- * In order to avoid this transport layer will offer a service called as Congestion control.
- * In congestion control, Our prime interest is to achieve average data rate
- * A peak data rate will ~~be~~ define the maximum bandwidth the channel is offering to the network users
- * In congestion, Maximum burst time is very crucial which refers to the maximum length at which the data is flowing with a peak speed.
- * We should also consider the bandwidth which the network should allocate for the flow of traffic.
- * There are 3 types of bit rates
 - 1) Constant bit rate (CBR)
 - 2) Variable bit rate (VBR)
 - 3) Bursty bit rate (BBR)



* Congestion will happen in any system where there is waiting in a network.

* In order to avoid this queuing policy is introduced which will ensure all the packets will be transmitted at their designated time intervals.

* Two issues need to be considered

a) If rate of the packet arrival is higher than the packet processing rate, then the input queue becomes longer.

b) If the packet departure is less than the processing rate, then the output queue will be longer.

* Congestion can be controlled using 2 categories

a) open loop congestion control

b) closed loop congestion control

Open loop congestion control

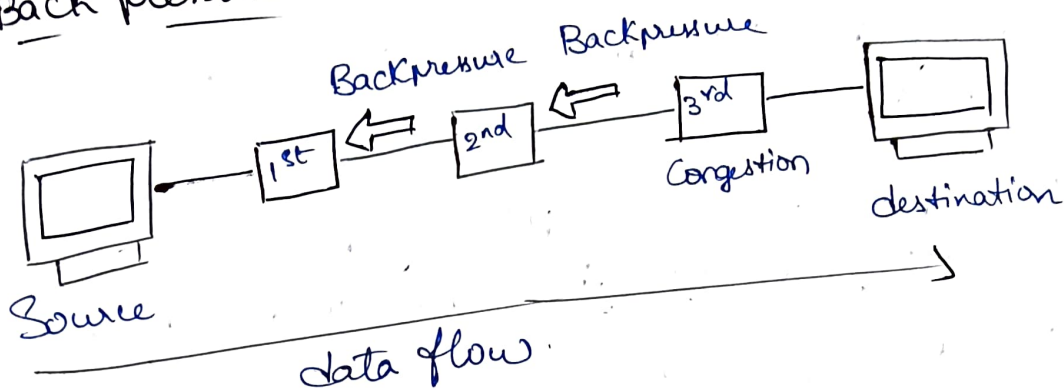
* In open loop: Retransmission policy, window policy, acknowledgement policy, discarding policy, admission policy

Closed loop congestion control

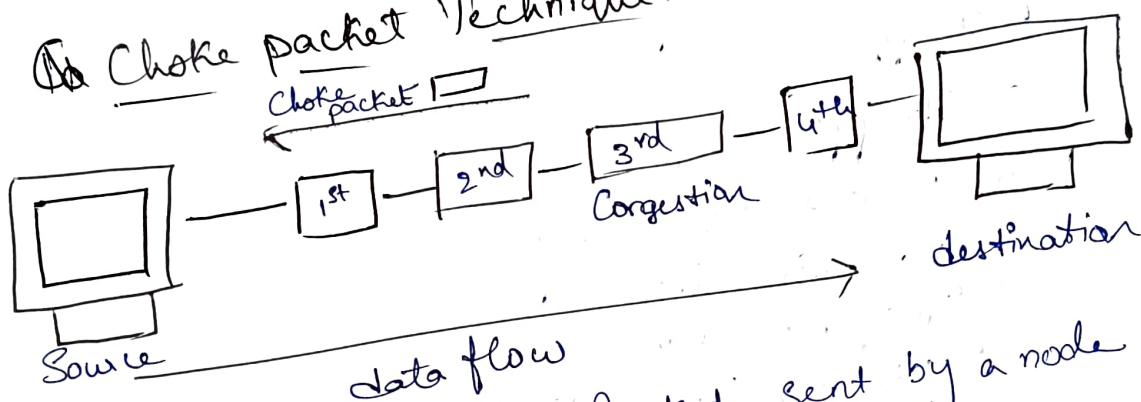
* In closed loop: Back pressure, choke packet, implicit signalling, explicit signalling.

21/10/24

Back pressure:



Choke packet Technique:



* A choke packet is a packet sent by a node to the source to inform it of congestion.

Implicit Signalling:

* In implicit Signalling, when a sender sends several packets, if there is no acknowledgement then the sender assumes, that there is congestion.

Explicit Signalling:

* In explicit Signalling, a signal is introduced along with the data to inform the sender about the congestion prevailing in the network.

* This signalling can be considered in both forward and backward directions.

* To improve the Quality in a network, transport layer has introduced 2 techniques such as leaky bucket, token bucket algorithms which can improve the flow of data.

* To improve the QoS, primary parameters such as reliability, delay, jitter, bandwidth are evaluated.

* QoS technique can use

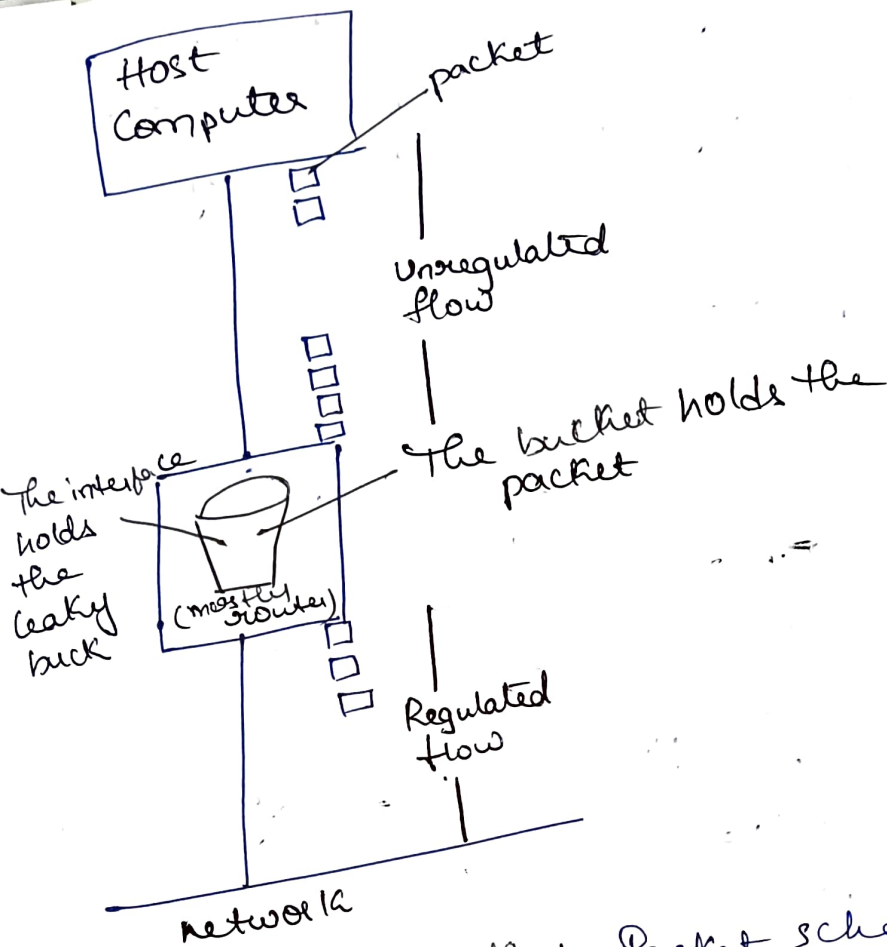
- 1) Overprovisioning
- 2) Buffering
- 3) Traffic shaping
- 4) Packet scheduling
- 5) Admission Control
- 6) Integrated services
- 7) Differentiated services.

1) Over provisioning: An easy solution is to provide ~~so~~ much router capacity.

2) Buffering: In buffering, the packets will be buffered and they will be asked to wait for some amount of time to use the network.

3) Traffic shaping: It is about regulating the average rate of data transmission. Traffic shaping reduces congestion. Monitoring a traffic flow is called traffic policing.
* In traffic shaping:

a) leaky bucket algorithm: No matter what the data rate is flowing, the outflow rate is always constant. The drawback is due to constant flow the data remaining is delayed (or) discarded.



c) Packet Scheduling: Packet scheduling deployed to avoid the overflow of data and it will introduce a fair queuing algorithm where every router will have separate queue for each output line one for each flow.

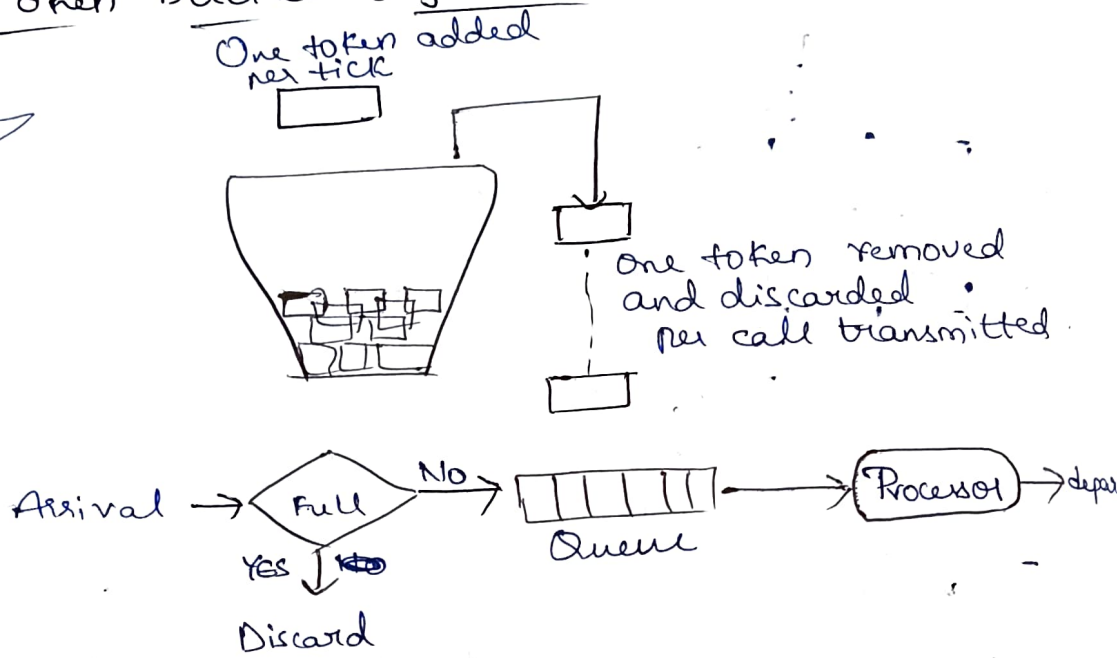
* When the output line becomes idle, the router will scan the queue taking the 1st packet on the next queue.

5) Admission Control:

Parameter	Unit
Token bucket rate	Bytes/sec
Token bucket size	Bytes
Peak data rate	Bytes/sec
Minimum Packet Size	Bytes
Maximum Packet Size	Bytes

* In admission policy control, token bucket algorithm is used, where the tokens are generated and given to the users and they can be utilized only at an appropriate data rate.

Token bucket algorithm



CO-4

Application layer:

- 1) DNS
- 2) SMTP
- 3) SNMP Introduction to Security
- 4) Security goals
- 5) Security Attacks
- 6) Security Services and Mechanisms
- 7) A Security Model
- 8) Asymmetric and Symmetric Key Ciphers
- 9) Substitution Techniques
- 10) Transposition Techniques
- 11) ~~DES~~ DES
- 12) RSA algorithm
- 13) Secure Socket Layer

28/10/24

CO-4
Application Layer

Simple Mail Transfer Protocol
(SMTP)

cc - carbon copy

bcc - blocked carbon copy

* In a Simple Mail Transfer Protocol, the protocol will define the template through which an email is to be sent across the network and the layout for the layout for the template is illustrated as follows.

To:	
cc:	
bcc:	
Subject:	
body:	
  	

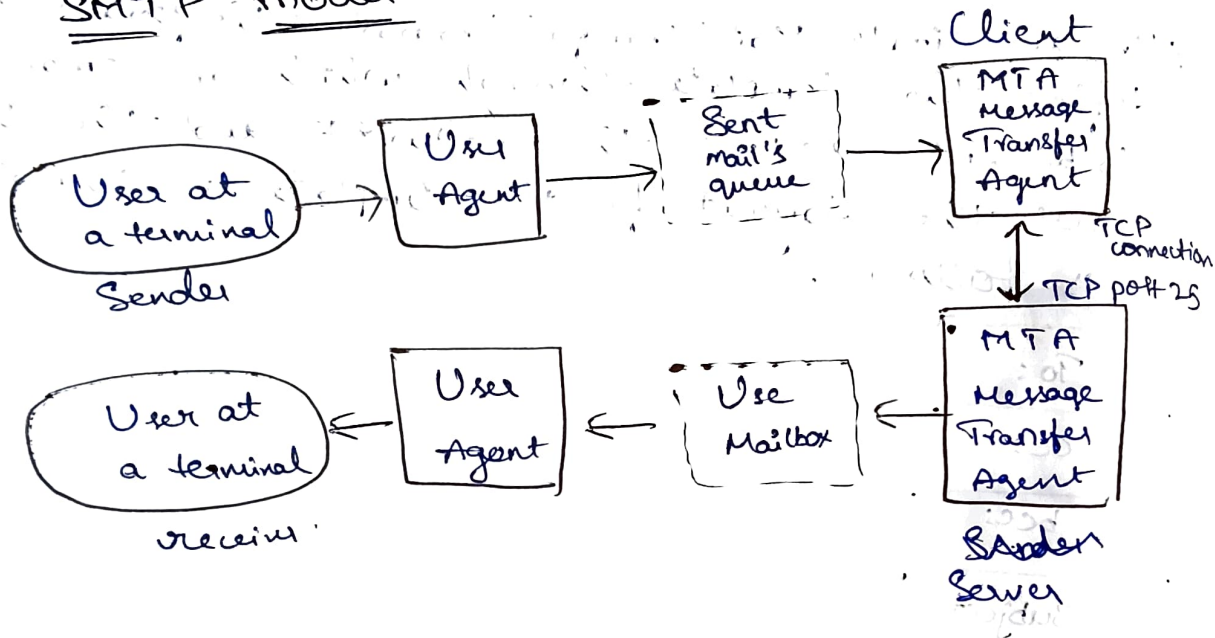
* SMTP follows 2 different methods to send an email.

- 1) End to End method
- 2) Store and forward method.

* In End to End method, when the users reside in different organizations, the emails will be sent instantly, where as if both the users are of same organisation, then store and forward method is used

* SMTP uses a TCP connection by establishing the link between sender and the receiver.

SMTP model



Components of SMTP

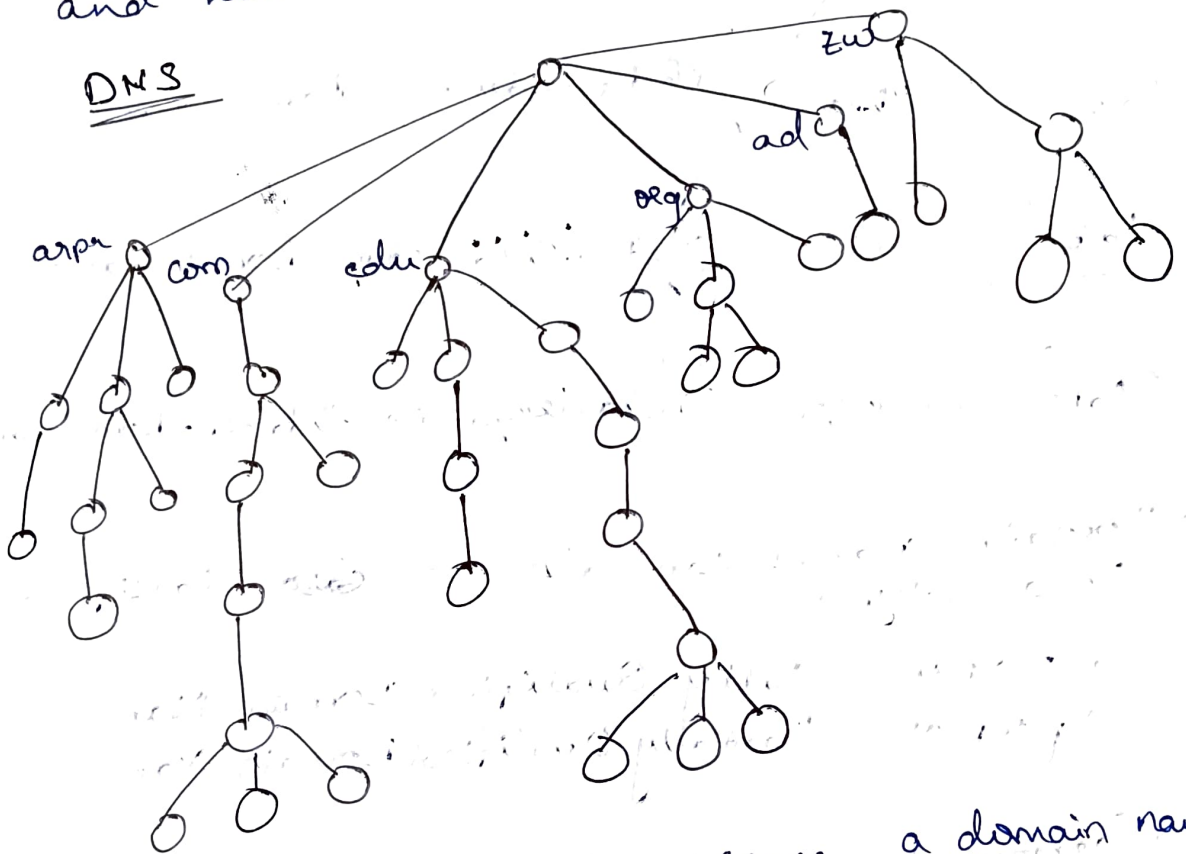
- 1) Mail User Agent (MUA)
- 2) Mail Submission Agent (MSA)
- 3) Mail Transfer Agent (MTA)
- 4) Mail Delivery Agent (MDA)

DNS (Domain Name Service)

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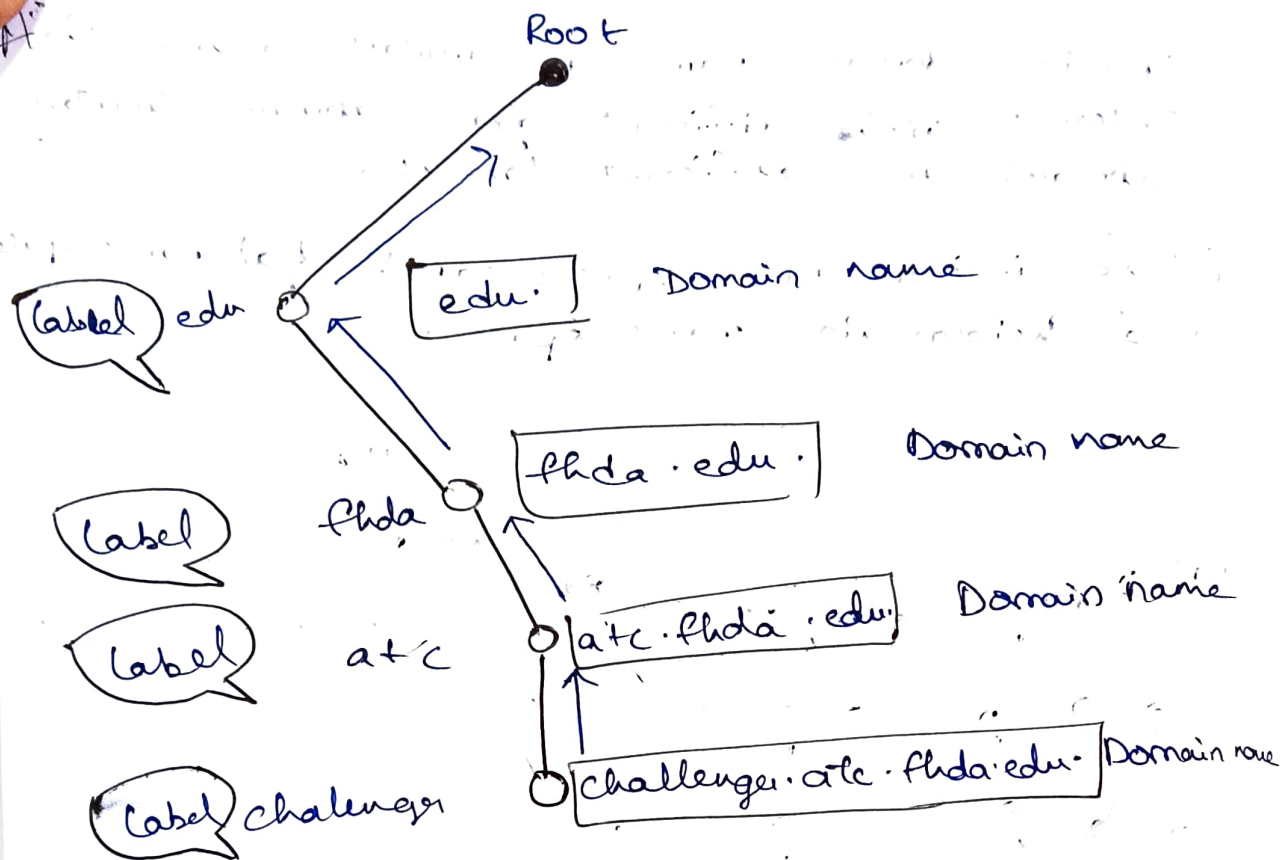
* Various domain names are available across the internet with different generic domains, country domains and different labels.

* The different name spaces can be flat name space and hierarchical name space.



* To have a hierarchical name space, a domain name space designed. In this design the names are defined in an inverted-tree structure with the root at top. The tree can have 0 to 127 (128) levels.

DOMAIN NAMES AND LABELS



* Domain names can also be ~~be~~ classified into 2 types.

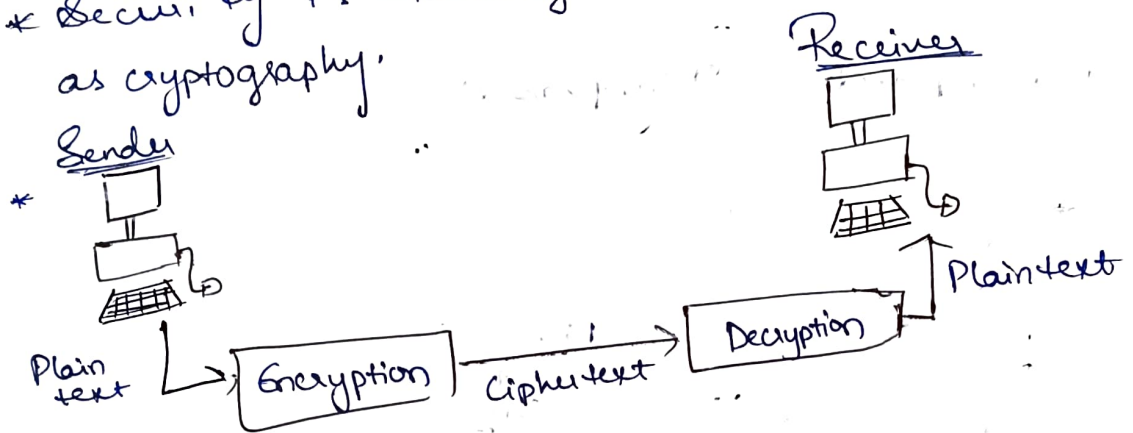
- 1) FQDN - Fully Qualified Domain Name
- 2) PQDN - Partially Qualified Domain Name

* Generic domain labels

- aero - aerospace (a) Airlines
- biz - business (b) firm
- com - commercial
- coop - cooperative business
- edu - education Institutions
- gov - Government Institutions.
- info - Information Service Providers
- int - International organizations
- mil - Military group.
- museum - museums
- name - Personal names
- net - Network support centres
- org - Non profit organizations
- pro - Professional Individual Organizations.

* Security needs to be provided to all the websites
 (a) the data which is communicated through the internet. So application layer along with providing the protocols should also provide a safe environment to send the data between the devices.

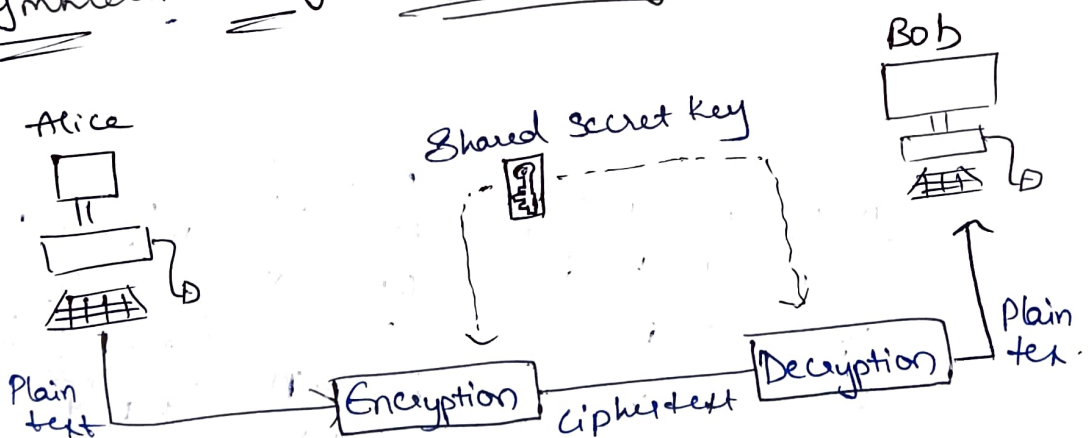
* Security is through the concept called as cryptography.



* Cryptography is divided into two types

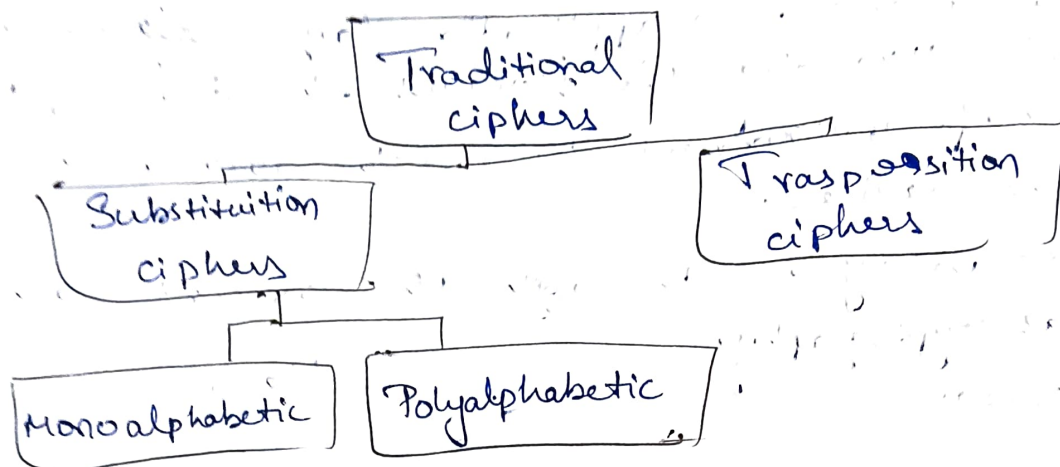
- a) Symmetric-key (Secret Key)
- b) Asymmetric-key (Public Key)

Symmetric Key Cryptography (Encryption)

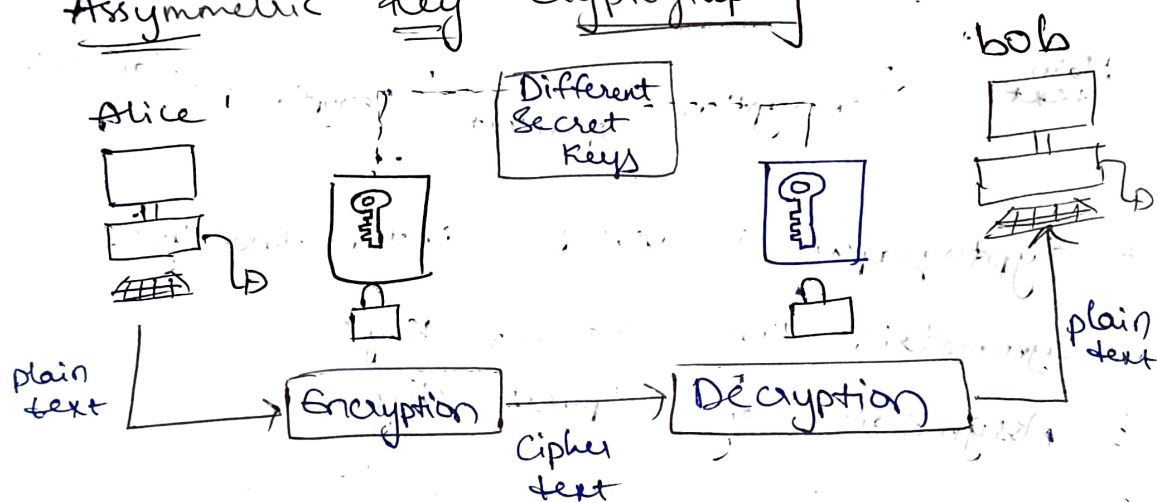


* In symmetric-key cryptography the same key is used by the sender and the receiver.

* Symmetric key cryptography is classified as



Asymmetric Key cryptography



Polyalphabetic:

m	o	n	a	r
c	h	y	b	d
i/j	k	l	p	q
s	t	u	v	w
x	z	e		

ne - mg
 tw - qz
 or - ~~an~~ nm
 kp - ft
 ro - mn
 to - pr
 co - hm
 ls - pt

m	o	n	a	r
c	h	y	b	d
e	f	g	l/j	k
t	p	q	s	u
v	v	w	x	z

6/11/24

Transposition Techniques

Ceaser Cipher

Plain text: Hello, world!

~~Key~~: Khoox, zroug (Encrypted)

Key: 3

$P = \text{encryption}(\text{plaintext}, \text{key})$

decryption value = ~~encrypt~~ (ciphertext, key) % 26

For encryption:

$$C = E(K, P) = (P + K) \% 26$$

For decryption:

$$P = D(K, C) = (C - K) \% 26$$

where, K takes on a value in the range 1 to 25.

Monoalphabetic: Every alphabet will have a replacement for the entire word.

Polyalphabetic:

Keyword: Monarchy

Plain text: Network Protocols

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	J
K	L	H P	Q	R S
T	U	V	W	X
Z				

ne - mg
tw -
or
kp
vo
to
co
ls

Keyword: Keyword

Plain text: Network protocols

K	e	y	w	o
r	d	a	b	c
f	g	h	i	j
m	n	p	q	s
t	u	v	x	z

ne - ud
tw - xk
on - kc
rp - ym
ro - ck
to - zk
co - lc
is - zs

Transposition techniques

Rail Fence Cipher

Step 1: The plain text is written as a sequence of diagonals.

Step 2: Then, to obtain cipher text, the text is read as a sequence of rows.

Example:

Let's say we take an example of

"INCLUDE HELP IS AWESOME"

2 rows - I C U E P S W S M N L D H L I A E O G

Cipher text - I C U E P S W S M N L D H L I A E O G

Columnar Transposition Technique

This technique is more complex as compared to the rail fence.

Step 1: In a rectangle of pre-defined size, write the plain-text message row by row.

Step 2: Read the plain message in random order in a column-wise fashion. It can be any order such as 2, 1, 3 etc.

Step 3: ~~the~~ Cipher text is obtained.

Keyword: Security

plain text: Network Protocols.

S	e	c	u	r
i	t	y	a	b
d	e	g	h	j
k	L	m	n	o
p	q	v	w	x/z

ne - ul
tw - aq
or - x b (a) zb
kp - ps
ro - bx(a) bz
to - bl
wo - rm
ls - ck

=> ulaqx bps bx blrmck

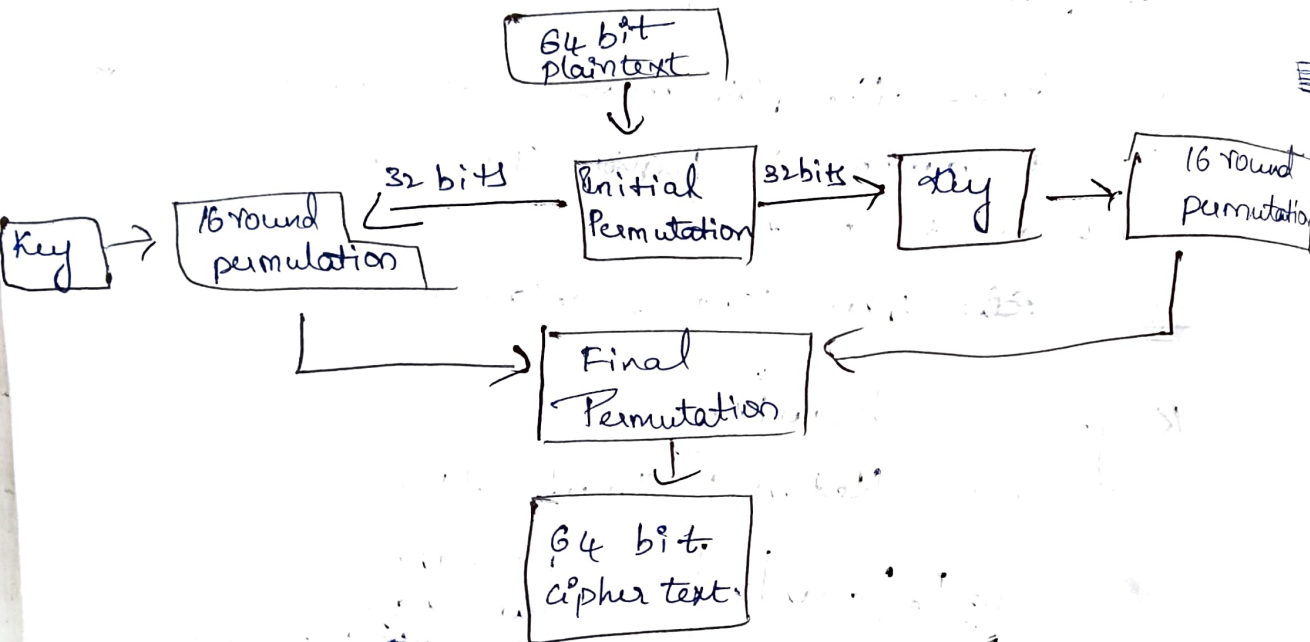
=> ul aq zb ps bz blrmck

12/11/24

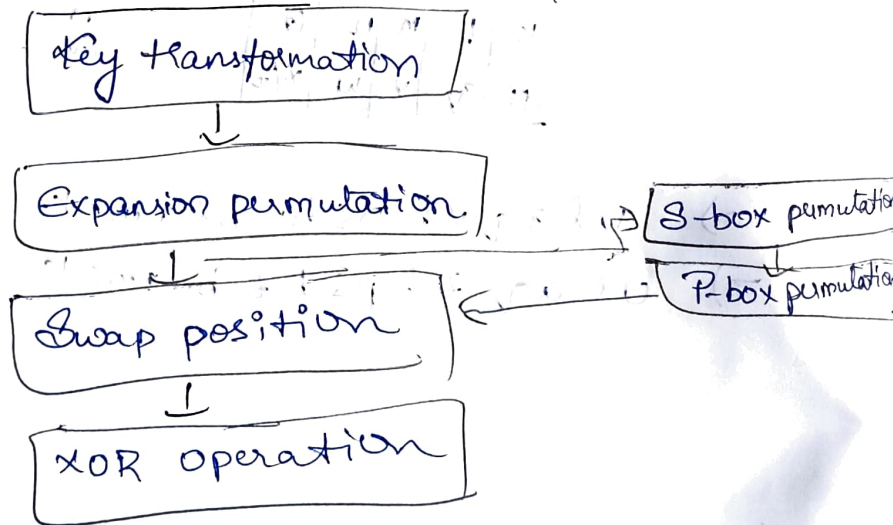
DES - Data Encryption Standard

* Data Encryption Technique, the plain text is 64 bit and key is 56 bits.

Broad Level Steps in DES

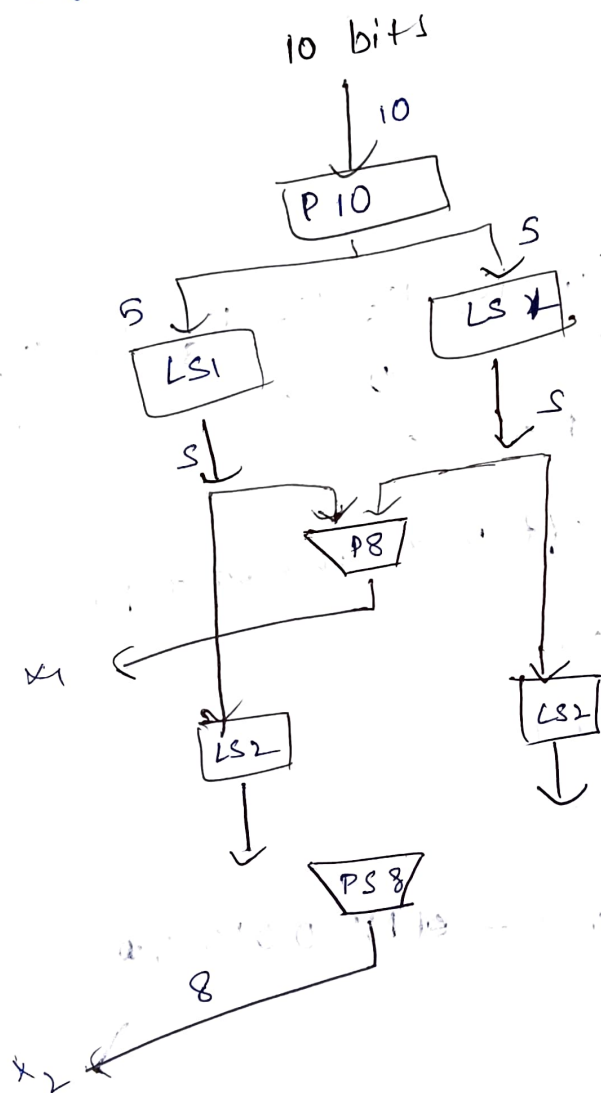


Steps for key generation



Simplified DES

*In a simplified DES the plain text is 8 bits
key is 10 bits,
key generation



Problem:

Given plain text and key for simplified DES is

10010111 - plain text

101000010 - key

3,5,2,7,4,10,1,9,8,6 - P10

6,3,7,4,8,5,10,9 - P8

2,4,3,1 - P4

2,6,3,1,4,8,5,7 - IP

4,1,3,5,7,2,8,6 - Inverse IP

4, 1, 2, 3, 2, 3, 4, 1 - Extended IP

S_0, S_1 - S boxes (which are also fixed and are defined as

$$S_0 = \begin{matrix} & 0 & 1 & 2 & 3 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix} \end{matrix}, S_1 = \begin{matrix} & 0 & 1 & 2 & 3 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix} \end{matrix}$$

Sol: (i) Key generation

$$\begin{array}{|c|c|c|c|c|c|c|c|c|c|} \hline K_1 & K_2 & K_3 & K_4 & K_5 & K_6 & K_7 & K_8 & K_9 & K_{10} \\ \hline 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline \end{array}$$

$P_{10} = 1000101100$ (3 5 2 7 4 10 1 9 8 6)
+ divide P_{10} into left part and right part.

$$\begin{array}{l} \rightarrow 10001 \\ \quad \underbrace{}_{\text{L.C. shift}} \\ \hline = 00001 \end{array} \quad \begin{array}{l} 01100 \\ \quad \underbrace{}_{\text{L.C. shift}} \\ \hline = 11000 \end{array}$$

$$P_8 = 637485109 - 0101000100$$

$$K_1 = 100100100$$

2-shift

$$0000111000$$

$$00100100011$$

$$K_2 = 0010000011$$

Key :- $K_1 \oplus K_2$
cipher text

Encryption Process:

plain text : 1 0 0 1 0 1 1 1

Initial permutation : 2 6 3 1 4 8 5 7

= 0 1 0 1 1 1 0 1

divide IP into two halves

0 1 0 1 | 1 1 0 1
L.P | R.P

Round 1:

* Take the right part of the plain text and perform extended permutation:

4	1	2	3	2	3	4	1
---	---	---	---	---	---	---	---

Right extended permutation : 1 1 1 0 1 0 1 1

* With the help of K_1 perform XOR operation and generate the values to be considered for the S-boxes.

$K_1 = 1 1 0 1 0 1 1 1$
 $0 1 0 0 0 1 0 0$

XOR $S_0 = 0 1 0 0$ $S_1 = 1 1 1 1$

* divide the given value into two halves

0 1 0 0 | 1 1 1 1

* for S0 box - 1, 4 bits $\rightarrow$ row
2, 3 bits $\rightarrow$ column

$S_0 = 0 0 = \text{row} = 0^{\text{th}}$
 $1 0 = \text{column} = 2^{\text{nd}}$
 $\Rightarrow 0^{\text{th}} \text{ row } 2^{\text{nd}} \text{ column} = 3 = 11$

$S_1 = 1 1 = 3$
 $1 1 = 3 \Rightarrow 3^{\text{rd}} \text{ row } 3^{\text{rd}} \text{ column} = 3 = 11 (\text{binary})$

$S_0 = 11, S_1 = 11 \Rightarrow 11 11$

Apply P_4 box to generate the new S values

$$P_4 = \begin{matrix} 2 & 4 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{matrix}$$

* Apply the Left part of the text and perform XOR operation

$$\begin{array}{r} P_4 = 1111 \\ C.P = 0101 \\ \hline 1010 \end{array}$$

* Now the new permuted text will be
 $1010 \mid 1101$ (Swap)

$$\Rightarrow 1101 \mid 1010$$

Round 2: $1101 \mid 1010$

* Take the ~~left~~^{right} part of permutation and perform extended permutation:

$$\begin{matrix} 4 & 1 & 2 & 3 & 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{matrix}$$

* With the help of K_2 perform XOR operation and generate the values that should be considered for S_0, S_1 boxes.

$$C.P: 01010101$$

$$K_2: 00000011$$

$$\text{XOR} \quad \begin{array}{r} 0001 \mid 0110 \\ \hline \end{array}$$

$$S_0 = \begin{matrix} 01 = 1 \\ 00 = 0 \end{matrix} \Rightarrow 1^{\text{st}} \text{ row } 0^{\text{th}} \text{ column} = 3 = 11$$

$$S_1 = \begin{matrix} 00 = 0 \\ 11 = 3 \end{matrix} \Rightarrow 0^{\text{th}} \text{ row } 3^{\text{rd}} \text{ column} = 3 = 11$$

$$P_4 = \begin{matrix} 2 & 4 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{matrix}$$

* Apply the Right part of the text and perform XOR operation!

P4: 1111

R.P: 1101
0010

= 0010 | 1010 = Inverse IP = 41357286
= 00111000

∴ Cipher text = 001011000

⇒ Perform SDES operation for the given plain
text - 1011101 for the key - 1010000010

Sol: ① key generation:

K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	K_{10}
1	0	1	0	0	0	0	0	1	0

P10: 3, 5, 2, 4, 4, 10, 1, 9, 8, 6
1, 0, 0, 0, 0, 0, 1, 1, 0, 0

10000
L.C shift

= 00001

01100
L.C shift

= 11000

P8 = 637488109 = 10101000

$K_1 = 1010$

K_2

12/11/24

RSA algorithm

- * Public key is used for encryption
- * Private key is used for decryption

Key Generation

Select p, q p and q both are prime, $p \neq q$

Calculate $n = p \times q$

Calculate $\phi(n) = (p-1) \times (q-1)$

Select integer e

$\bullet \gcd(\phi(n), e) = 1$

Calculate d

$1 < e < \phi(n)$
 $d = e^{-1} \pmod{\phi(n)}$

Public key (e)

$pu = (e, n)$

Private key (d)

$pv = (d, n)$

- * Take a relative prime number to calculate $\gcd$ and the conditions to be $1 < e < \phi(n)$.

Encryption Calculation (done with public key)
of Plain text (m)

$$\text{Cipher text } (C) = m^e \pmod{n}$$

of Decryption calculation the (done with private key)

$$\text{plain text } (m) = C^d \pmod{n}$$

- * Select 2 prime numbers $p=53$, $q=59$ compute the public key and the private key for the given Plain text "HI"

Sol: $n = p \times q = 53 \times 59 = \del{74} 3127$

$$\phi(n) = (p-1) \times (q-1) = 3016$$

$$e=2 \Rightarrow 1 < e < \phi(n) = 1 < 2 < 3016 \text{ True}$$

$$\gcd(e, \phi(n)) = (3, 3016) = 1 \quad \text{True}$$

for 'e' compute $\gcd(\phi(n), e) = 1$
 $1 < e < \phi(n)$

$$d = e^{-1} \bmod \phi(n)$$

$$= \frac{1}{3} \% 3016$$

$$e = 3, d = 2011$$

$$C = 89^3 \bmod 3127 = 1394$$

$$M = 1394^{2011} \bmod 3127 =$$

13/11/24

Problem

$$p = 3, q = 5, M = 8$$

Sol:

$$n = p \times q = 3 \times 5 = 15$$

$$\phi(n) = (p-1) \times (q-1) = 2 \times 4 = 8 \quad \text{— prime numbers less than } \phi(n) = 1, 3, 5, 7$$

$$\gcd(e, \phi(n)) = 1 \text{ and } 1 < e < \phi(n) \text{ — true}$$

$\begin{matrix} \gcd(e, 3) = 1 \\ e = 3 \end{matrix}$

$$\times \begin{pmatrix} d = e^{-1} \bmod \phi(n) \\ d = \frac{1}{3} \bmod 8 \end{pmatrix} \times$$

$$d \times e \bmod \phi(n) = 1$$

$$d \times 3 \bmod 8 = 1$$

$$d = 3 \Rightarrow 3 \times 3 \bmod 8 = 1$$

$$9 \bmod 8 = 1$$

$$\Rightarrow \boxed{d = 3} \quad \boxed{e = 3} \quad \boxed{n = 15} \quad \boxed{\phi(n) = 8}$$

$$C = 8^3 \bmod 15 = 2$$

$$M = 2^3 \bmod 15 = 8 \bmod 15 = 8$$

$$p = 5, q = 7, P = 8$$

sol: $n = p \times q = 5 \times 7 = 35$

$$\phi(n) = (p-1) \times (q-1) = 4 \times 6 = 24$$

$$\gcd(e, \phi(n)) = 1 \quad \& \quad 1 < e < 24$$

$$e = 5$$

$$d \times e \bmod \phi(n) = 1$$

$$d \times 5 \bmod 24 = 1$$

$$d = 5$$

$$\boxed{e = 5} \quad \boxed{d = 5} \quad \boxed{n = 35} \quad \boxed{\phi(n) = 24}$$

$$C = 8^{n^5} \bmod 35 = \cancel{250} 8$$

$$P = 8^{n^5} \bmod 24 = 8$$

$$p = 17, q = 11, P = 89$$

sol:

$$n = 17 \times 11 = 187$$

$$\phi(n) = 16 \times 10 = 160$$

$$\gcd(e, \phi(n)) = 1 \quad \& \quad 1 < e < \underline{160}$$

$$e = \underline{7}$$

$$d \times e \bmod 160 = 1$$

$$d \times 7 \bmod 160 = 1$$

$$d = \cancel{59} 23$$

$$\boxed{d = 23} \quad \boxed{e = 7} \quad \boxed{n = 187} \quad \boxed{\phi(n) = 160}$$

$$C = 89^{n^7} \bmod \cancel{187} = \cancel{184} 8$$

$$M = 184^{n^{23}} \bmod 187 = 73$$